Power path on every rail: RAW -> eFuse -> INA260 VIN+ -> INA260 VIN- -> PROTECTED output. INA260 VIN- is the downstream positive rail, never GND.

4-layer PCB intent: L1 components/high-current/local signals, L2 solid GND reference, L3 power zones, L4 secondary routing/GND fill.

Serviceable layout choice: eFuse ICs are on the main PCB for manufacturer assembly. For true field replacement, move each eFuse circuit to a plug-in daughterboard.

Only user-selected H1/H2 mission power and GND pins are connected. Excluded communication/normally-closed pins are not represented here.

ATX input and control terminals

J1
+3V3_RAW 1 Pin_1
+3V3_RAW 2 Pin_2
+5V_RAW 3 Pin_3
+5V_RAW 4 Pin_4
+12V_RAW 5 Pin_5
+12V_RAW 6 Pin_6
GND 7 Pin_7
GND 8 Pin_8
GND 9 Pin_9
GND 10 Pin_10
J_ATX_PWR: 3V3 3V3 5V 5V 12V 12V GNDx4

J2
ATX_PS_ON 1 Pin_1
ATX_PWR_OK 2 Pin_2
+5VSB 3 Pin_3
GND 4 Pin_4
J_ATX_CTRL: PS_ON PWR_OK 5VSB GND

J3
ATX_PS_ON 1 Pin_1
GND 2 Pin_2
ATX_SWITCH

Protected rail chains

3.3V rail: RAW/source -> eFuse -> INA260 VIN+ / VIN- -> protected bus

R2 +3V3_RAW 1 2 RESET_3V3 — 100k 1% EN top 2.40V
RESET_3V3 1 2 GND — 100k 1% EN bottom
R4 +3V3_RAW 1 2 OVLO_3V3 — 24.9k 1% OVLO top 4.19V
R5 OVLO_3V3 1 2 GND — 10k 1% OVLO bottom

U1
RESET_3V3 1 EN/UVLO
OVLO_3V3 2 EN/OVLO
+3V3_RAW 3 VIN
GND 8 GND
ILIM_3V3 5 ILIM
PG 3 PG_3V3
-FLT 4 FLT_3V3
OUT 6 +3V3_EFUSE_OUT
DVDT_OPEN 7
ITIMER_OPEN 10
TPS259814L 3V3 eFuse latch-off

C1 +3V3_RAW 1 2 GND — 1uF 25V X7R input close to eFuse
C2 +3V3_EFUSE_OUT 1 2 GND — 10uF 25V X7R output close to eFuse
R1 ILIM_3V3 1 2 GND — 1.10k 1% ILIM ~6A

U5
+3V3_EFUSE_OUT VIN+
+ARDUINO_5V VCC
GND 2 GND
VIN- 8 +3V3_PROTECTED
SCL 3 I2C_SCL
SDA 4 I2C_SDA
ALERT 5 ALERT_3V3
VBUS_NC 6
INA260 3V3 module

+3V3_PROTECTED branches only after INA260 VIN-

5V rail: RAW/source -> eFuse -> INA260 VIN+ / VIN- -> protected bus

R10 +5V_RAW 1 2 UVLO_5V — 261k 1% UVLO top 4.33V
R11 UVLO_5V 1 2 GND — 100k 1% UVLO bottom
R12 +5V_RAW 1 2 OVP_5V — 383k 1% OVP top 5.80V
R13 OVP_5V 1 2 GND — 100k 1% OVP bottom

U2
+5V_RAW 1 VIN
+5V_RAW 2 VIN
+5V_RAW 3 VIN_SYS
UVLO 5 UVLO
OVP 6 OVP
ILIM 7 ILIM
RESET 12 ~SHDN
MODE_OPEN 11
GND 8 GND
GND 25 EP_GND
OUT 17 +5V_EFUSE_OUT
OUT 18 +5V_EFUSE_OUT
-FLT 14 FLT_5V
PGOOD 16 PG_5V
PGTH 15
IMON 13 IMON_5V
dVdT_OPEN 9
B_GATE_OPEN 3
DRV_OPEN 4
TPS26631RGE 5V eFuse

C3 +5V_RAW 1 2 GND — 1uF 25V X7R input close to eFuse
C4 +5V_EFUSE_OUT 1 2 GND — 10uF 25V X7R output close to eFuse
R33 ATX_PS_ON 1 2 GATE_PS_ON — 330 gate
R34 GATE_PS_ON 1 2 GND — 100k gate pulldown
R9 ILIM_5V 1 2 GND

U6
+5V_EFUSE_OUT VIN+
+ARDUINO_5V VCC
GND 2 GND
VIN- 8 +5V_PROTECTED
SCL 3 I2C_SCL
SDA 4 I2C_SDA
ALERT 5 ALERT_5V
VBUS_NC 6
INA260 5V module

+5V_PROTECTED branches only after INA260 VIN-

12V rail: RAW/source -> eFuse -> INA260 VIN+ / VIN- -> protected bus

Q5 2N7002 ATX PS_ON# pulldown: GATE_PS_ON 1, ATX_PS_ON_N 3, GND 2
R18 +12V_RAW 1 2 UVLO_12V — 806k 1% UVLO top 10.87V
R19 UVLO_12V 1 2 GND — 100k 1% UVLO bottom
R20 +12V_RAW 1 2 OVP_12V — 1.05M 1% OVP top 13.80V
R21 OVP_12V 1 2 GND — 100k 1% OVP bottom

U3
+12V_RAW 1 VIN
+12V_RAW 2 VIN
+12V_RAW 3 VIN_SYS
UVLO 5 UVLO
OVP 6 OVP
ILIM 7 ILIM
RESET 12 ~SHDN
MODE_OPEN 11
GND 8 GND
GND 25 EP_GND
OUT 17 +12V_EFUSE_OUT
OUT 18 +12V_EFUSE_OUT
-FLT 14 FLT_12V
PGOOD 16 PG_12V
PGTH 15
IMON 13 IMON_12V
dVdT_OPEN 9
B_GATE_OPEN 3
DRV_OPEN 4
TPS26631RGE 12V eFuse

U7
+12V_EFUSE_OUT VIN+
+ARDUINO_5V VCC
GND 2 GND
VIN- 8 +12V_PROTECTED
SCL 3 I2C_SCL
SDA 4 I2C_SDA
ALERT 5 ALERT_12V
VBUS_NC 6
INA260 12V module

+12V_PROTECTED branches only after INA260 VIN-

Boost module: verified 60 mm x 42 mm x 20 mm footprint, +12V in and 16.8V out

U9
GND 1 IN-
+12V_RAW 2 IN+
OUT- 3 GND
OUT+ 4 +16V8_RAW
SW1_OPT 5
SW2_OPT 6
Boost 12V to 16.8V module 60x42 VERIFIED

C5 +12V_RAW 1 2 GND — 1uF 50V X7R input close to eFuse
C6 +12V_EFUSE_OUT 1 2 GND — 10uF 50V X7R output close to eFuse

16.8V rail: RAW/source -> eFuse -> INA260 VIN+ / VIN- -> protected bus

R26 +16V8_RAW 1 2 UVLO_16V8 — 1.07M 1% UVLO top 14.04V
R27 UVLO_16V8 1 2 GND — 100k 1% UVLO bottom
ILIM_16V8 1 2 GND — 3.00k 1% ILIM ~6A
R28 +16V8_RAW 1 2 OVP_16V8 — 1.40M 1% OVP top 18.00V
R29 OVP_16V8 1 2 GND — 100k 1% OVP bottom

U4
+16V8_RAW 1 VIN
+16V8_RAW 2 VIN
+16V8_RAW 3 VIN_SYS
UVLO 5 UVLO
OVP 6 OVP
ILIM 7 ILIM
RESET 12 ~SHDN
MODE_OPEN 11
GND 8 GND
GND 25 EP_GND
OUT 17 +16V8_EFUSE_OUT
OUT 18 +16V8_EFUSE_OUT
-FLT 14 FLT_16V8
PGOOD 16 PG_16V8
PGTH 15
IMON 13 IMON_16V8
dVdT_OPEN 9
B_GATE_OPEN 3
DRV_OPEN 4
TPS26631RGE 16V8 eFuse

U8
+16V8_EFUSE_OUT VIN+
+ARDUINO_5V VCC
GND 2 GND
VIN- 8 +16V8_PROTECTED
SCL 3 I2C_SCL
SDA 4 I2C_SDA
ALERT 5 ALERT_16V8
VBUS_NC 6
INA260 16V8 module

+16V8_PROTECTED branches only after INA260 VIN-

C7 +16V8_RAW 1 2 GND — 1uF 50V X7R input close to eFuse
C8 +16V8_EFUSE_OUT 1 2 GND — 10uF 50V X7R output close to eFuse
R25 ILIM_16V8 1 2 GND — 3.00k 1% ILIM ~6A

Output connectors: dedicated terminal J4 plus selected EPS H1/H2 power/GND pins only

R8 FLT_3V3 1 2 +ARDUINO_5V — 10k FLT pullup
R24 FLT_5V 1 2 +ARDUINO_5V — 10k FLT pullup
R32 FLT_16V8 1 2 +ARDUINO_5V — 10k FLT pullup
R16 FLT_5V 1 2 +ARDUINO_5V — 10k FLT pullup

J4
+3V3_PROTECTED 1 Pin_1
GND 2 Pin_2
+5V_PROTECTED 3 Pin_3
GND 4 Pin_4
+12V_PROTECTED 5 Pin_5
GND 6 Pin_6
+16V8_PROTECTED 7 Pin_7
GND 8 Pin_8
DEDICATED OUTPUTS: 3V3/G 5V/G 12V/G 16V8/G

JH1
GND 22 H1_22
H1_24 24 GND
GND 27 H1_27
H1_31 31 GND
GND 32 H1_32
H1_36 36 GND
GND 39 H1_39
H1_40 40 GND
GND 42 H1_42
H1_47 47 +5V_PROTECTED
+3V3_PROTECTED H1_48
H1_49 49 +5V_PROTECTED
+12V_PROTECTED H1_52
H1 selected power/GND pins only

JH2
+5V_PROTECTED H2_25
H2_26 26 +5V_PROTECTED
+3V3_PROTECTED H2_27
H2_28 28 +3V3_PROTECTED
GND 29 H2_29
H2_30 30 GND
GND 31 H2_31
H2_32 32 GND
GND 36 H2_36
H2_39 39 GND
GND 40 H2_40
H2_41 41 GND
GND 42 H2_42
H2_43 43 GND
GND 44 H2_44
H2_45 45 +16V8_PROTECTED
+16V8_PROTECTED H2_46
H2 selected power/GND pins only

Test points

TP1 +3V3_RAW
TP2 +3V3_EFUSE_OUT
TP3 +3V3_PROTECTED
TP4 +5V_RAW
TP5 +5V_EFUSE_OUT
TP6 +5V_PROTECTED
TP7 +12V_RAW
TP8 +12V_EFUSE_OUT
TP9 +12V_PROTECTED
TP10 +16V8_RAW
TP11 +16V8_EFUSE_OUT
TP12 +16V8_PROTECTED
TP13 GND 1
GND
TP14 I2C_SDA
I2C_SDA
TP15 I2C_SCL
I2C_SCL
TP16 ATX_PS_ON_N
ATX_PS_ON_N
TP17 ATX_PWR_OK
ATX_PWR_OK
TP18 +5VSB
+5VSB
TP19 RESET_3V3
RESET_3V3
TP20 RESET_5V
RESET_5V
TP21 RESET_12V
RESET_12V
TP22 RESET_16V8
RESET_16V8
TP23 FLT_3V3
FLT_3V3
TP24 FLT_5V
FLT_5V
TP25 FLT_12V
FLT_12V
TP26 FLT_16V8
FLT_16V8
TP27 +12V_RAW
BOOST_IN+ (+12V_RAW)
TP28 +16V8_RAW
BOOST_OUT+ (+16V8_RAW)
TP29 GND 1
BOOST_GND

Arduino/control section: USB-powered Nano, open-drain reset MOSFETs, FLT inputs, PS_ON# pulldown

A1
D1/TX 1
D0/RX 2
GND 4
ARD_D3 D3
ARD_D4 D4
ARD_D5 D5
ARD_D6 D6
FLT_5V D7
FLT_3V3 D8
FLT_12V D9
FLT_16V8 D10
ATX_PS_ON D11
ATX_PWR_OK D12
5V 27 +ARDUINO_5V
GND 29 GND
A4/SDA 23 I2C_SDA
A5/SCL 24 I2C_SCL
Arduino Nano USB-powered socket

Edge-access reset buttons: RESET_3V3, RESET_5V, RESET_12V, RESET_16V8

R6 ARD_D3 1 2 GATE_3V3 — 330 gate
R7 GATE_3V3 1 2 GND — 100k gate pulldown
R14 ARD_D4 1 2 GATE_5V — 330 gate
R15 GATE_5V 1 2 GND — 100k gate pulldown
R22 ARD_D5 1 2 GATE_12V — 330 gate
R23 GATE_12V 1 2 GND — 100k gate pulldown
R30 ARD_D6 1 2 GATE_16V8 — 330 gate
R31 GATE_16V8 1 2 GND — 100k gate pulldown

Q1 2N7002 RESET_3V3 pulldown: GATE_3V3 1, RESET_3V3 3, GND 2
Q2 2N7002 RESET_5V pulldown: GATE_5V 1, RESET_5V 3, GND 2
Q3 2N7002 RESET_12V pulldown: GATE_12V 1, RESET_12V 3, GND 2
Q4 2N7002 RESET_16V8 pulldown: GATE_16V8 1, RESET_16V8 3, GND 2

SW1 RESET_3V3 1 2 GND — RESET_3V3 edge button
SW2 RESET_5V 1 2 GND — RESET_5V edge button
SW3 RESET_12V 1 2 GND — RESET_12V edge button
SW4 RESET_16V8 1 2 GND — RESET_16V8 edge button